

FLAVOURBENCH: Ranking Frontier Language Models with Executable Culinary Ground Truth

Josef Chen

Independent Researcher

Erim Hayretci

Imperial College London

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Abstract

Open-ended language-model benchmarks usually inherit a judge: a human preference panel, another model, or a brittle exact-match key. We introduce FLAVOURBENCH, an automated benchmark in which a versioned culinary system supplies dense, executable ground truth. Each task presents eight ingredients and asks for a three-ingredient portfolio; before model execution, EPICURE scores all 56 possible portfolios. We evaluate 27 frontier endpoints on an identical 534-task core spanning substitution, pairing, and constrained composition. Every ranked model has exactly 89 valid responses per panel and family (14418 model-task cells total), eliminating differential missingness from the leaderboard. The FlavourBench Score is the equal-family mean of the frozen task scores. We use 50000 anchor-cluster bootstrap replicates for simultaneous 95% score bands and 100000 sign-flip draws for all 351 paired model contrasts, with Holm control. The two independently compiled panels correlate at $r = 0.89$ (rank $\rho = 0.80$), and crossed-design relative-decision generalizability is 0.936. Grok 4.6 has the largest point estimate at 65.1 (simultaneous 95% CI 61.0–69.2); 101 of 351 model pairs are resolved. The release includes the prompts, all portfolio score maps, raw responses, exact routes, content hashes, and an offline verifier that reconstructs every result.

1 Introduction

Benchmarks are strongest when the evaluator can be executed. Code can be tested, database states can be inspected, and games can be scored. Culinary reasoning is usually evaluated with a model judge, a handful of human preferences, or factual questions whose answers do not capture the quality of a complete decision. That leaves two problems: the judge is entangled with the systems being tested, and exact match throws away useful differences between plausible answers.

FLAVOURBENCH turns a culinary runtime into a benchmark environment. EPICURE represents 1,790 ingredients in a 300-dimensional space and exposes deterministic operations for substitution, pairing, dietary feasibility, and regional composition [12, 13]. A task compiler converts these operations into three-of-eight selection problems. It enumerates every candidate portfolio and freezes a continuous 0–100 score map before model execution. An answer is therefore scored by table lookup, without a model judge and without post-hoc interpretation.

This setup separates the reference system from the evaluated systems. EPICURE is not a leaderboard entry and is not “100% accurate.” It defines the task-specific reward surface. A model receives 100 on a task only if it selects that task’s highest-scoring portfolio; other valid portfolios receive partial credit. The benchmark measures agreement with a published executable culinary environment, not universal human taste.

We use that environment to evaluate 27 current endpoints. Every model faces the same 534 tasks and 56 frozen answer scores per task. The complete common core supports direct paired comparisons on 534 shared

observations per model pair instead of inferring a ranking from sparse votes or comparing endpoints on different task subsets.

The paper makes three contributions:

1. an executable culinary benchmark with 29904 model-independent portfolio scores and a single interpretable 0–100 metric;
2. a 27-model, 534-task complete-core evaluation with simultaneous uncertainty and paired, multiplicity-controlled tests; and
3. a content-addressed release in which every public number traces to frozen tasks, exact routes, raw responses, and a dependency-light offline scorer.

2 Executable culinary ground truth

2.1 A reference environment, not a model judge

The benchmark binds an exact EPICURE release, evidence bundle, application build, ingredient count, and embedding dimensionality by hash. Changing any of these inputs creates a new benchmark task-set identity. This is analogous to fixing a simulator or a code-test suite: the reference is explicit and reproducible, but its external validity remains a scientific question rather than an assumption hidden inside a judge prompt.

For each task t , let $\mathcal{A}_t$ contain all 56 three-item subsets of eight candidates. EPICURE assigns a raw utility $u_t(a)$ to every valid $a \in \mathcal{A}_t$. Invalid portfolios under a dietary or category constraint receive zero. Valid utilities are

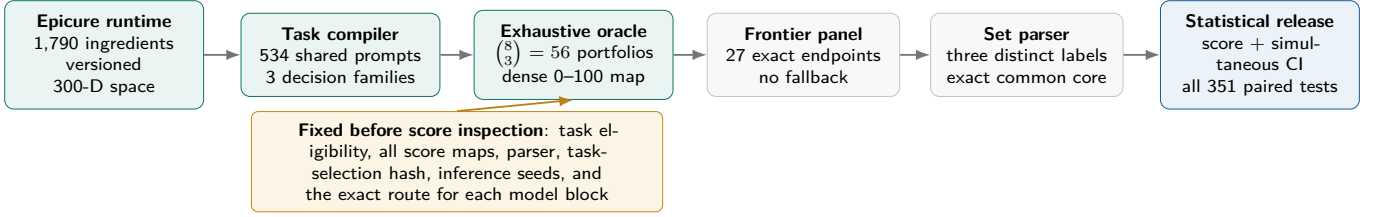


Figure 1: FLAVOURBENCH compiles a versioned culinary runtime into dense selection tasks. The evaluated models never judge one another and do not access EPICURE. All score maps and the per-panel contracts are fixed before collection. The complete-core task set is selected from response validity alone, before inspecting any model selection or score.

normalized within the task:

$$s_t(a) = 100 \frac{u_t(a) - \min_{b \in \mathcal{A}_t} u_t(b)}{\max_{b \in \mathcal{A}_t} u_t(b) - \min_{b \in \mathcal{A}_t} u_t(b)}. \quad (1)$$

Every task has a unique 100-point optimum. The exact chance baseline is not a generic percentage; it is $|\mathcal{A}_t|^{-1} \sum_a s_t(a)$ for that frozen task.

2.2 Three complete-core decision families

The ranked core contains 534 tasks: 89 from each family in each of two independently compiled panels, spanning 534 unique anchor ingredients. Tasks sharing an anchor move together as one cluster in inference. Candidate sets are selected across validation strata, then ordered and labeled by a task-derived hash. No evaluated model writes an item, distractor, or score. Regional-composition tasks are released as a supplemental track but are not in the 27-model leaderboard because they do not support an equally complete common core.

The score maps have substantial resolution rather than one correct key plus 55 equivalent mistakes. Table 2 reports the exact random-portfolio baseline, separation of the best and second-best portfolios, and the number of distinct score values. Constraint tasks are intentionally sparse because infeasible portfolios score zero; the other families provide much finer graded rewards.

2.3 Prompt and parser contract

Prompts show the decision context and eight labeled candidates. The final line must begin with the `FINAL_SELECTION` marker and contain three distinct A–H labels separated by commas. No prompt supplies a concrete valid triplet. The parser takes the final exact marker line, uppercases the labels, sorts them as an unordered set, and performs one lookup in the frozen score map. Explanatory text does not affect the result. Missing markers, duplicate labels, provider failures, and abnormal finishes remain in the raw response ledger but cannot enter the common core.

The model-only condition is deliberate. EPICURE supplies the ground truth but is not exposed as a tool during ranking. Thus the score measures whether a model can

make the culinary decision, not whether it can copy a runtime return value after being told which operation to call.

3 Evaluation design

3.1 One primary score

For model m , panel p , family f , and task t , let Y_{mt} be the frozen portfolio score and let T_{pf} be the score-blindly selected set of 89 tasks completed by every model. The FlavourBench Score is

$$F_m = \frac{1}{2 \cdot 3} \sum_{p=1}^2 \sum_{f=1}^3 \frac{1}{|T_{pf}|} \sum_{t \in T_{pf}} Y_{mt}. \quad (2)$$

It ranges from 0 to 100 and equals the ordinary mean over all 534 cells because the six panel–family strata have equal size. Every ranked endpoint contributes exactly 534 valid responses. Failures are neither assigned zero nor silently dropped model by model: they determine whether a task can enter the shared core, using response status and parser validity only. The task-selection hash is fixed before loading any selected portfolio or score.

3.2 Frontier endpoint panel

The panel spans OpenAI, Anthropic, Google, xAI, Meta, Kimi, Qwen, Z.ai, DeepSeek, ByteDance, Thinking Machines, MiniMax, NVIDIA, Mistral, Tencent, and Cohere. It includes GPT-5.6 Sol, Terra, and Luna; Claude Opus, Sonnet, and Fable 5; Gemini 3.1 Pro and 3.6 Flash; Grok 4.6; Llama 4 Maverick; Muse Spark 1.2 and Muse Glimmer 30B; Kimi K3; Qwen 3.8 Max and Qwen3.8 2.4T A95B; GLM 5.2 and GLM 5.3; DeepSeek V4 Pro 0813 and V4 Flash; Seed 2.1 Turbo; Inkling; MiniMax M3; Nemotron 3.5 Lightning; Mistral Large 3; Tencent HY 3; Cohere Command A; and Cohere Command R+ (08-2024). Z.ai supplied written permission for one finite GLM 5.3 benchmark run, conditioned on the adapter not becoming a permanent running function. We bind that permission and its hash in the release, require the returned identity to equal `glm-5.3`, and expose no standing endpoint.¹

¹<https://docs.z.ai/guides/llm/glm-5.3>

Family	Decision	Epicure utility	Hard constraint	Tasks
Substitution	choose a three-item replacement portfolio for an anchor ingredient	0.8 anchor similarity +0.2 portfolio coherence	none	178
Pairing	choose three ingredients to accompany an anchor	0.65 anchor affinity +0.35 portfolio coherence	none	178
Constraints	choose a feasible portfolio under diet and processing limits	0.7 anchor similarity +0.3 coherence	diet and maximum NOVA level	178

Table 1: The three complete-core task families share one response and scoring interface but probe different culinary decisions. Each task contains eight candidates and 56 frozen scores.

Family	Exact chance	Top gap	Distinct scores
Substitution	45.2	5.6	56
Pairing	45.0	5.6	56
Constraints	3.4	37.3	4

Table 2: Frozen task-map diagnostics for 178 scheduled tasks per family, computed before model execution. Chance is the uniformly random portfolio mean; top gap is the median best-second-best margin.

Routes were chosen before each scored block and automatic fallback is disabled. Most endpoints use an exact OpenRouter route; Qwen 3.8 Max uses Alibaba, Fable 5 uses Google Vertex, GLM 5.3 uses the finite Z.ai route above, and selected DeepSeek blocks use an exact GMI-Cloud or DeepInfra route. When transport validation required a replacement, the complete model block was rerun under one new route; successful cells from superseded blocks are not pooled. The manifest binds the requested model, returned identity contract, endpoint tag, provider, supported parameters, and execution contract for both panels. The exact per-model routes appear in Appendix A. For endpoints that advertise a reasoning-effort control, hidden reasoning is excluded and effort is fixed to the lowest transport-stable setting recorded in the exact analysis plan. Endpoints without that control use their provider-fixed behavior. This prevents hidden deliberation from silently consuming the answer budget while leaving the scored response format identical across routes.

3.3 Score-blind complete-core construction

The two task compilers each produce 160 candidates per family. After collection, the release re-scores response syntax with one versioned parser and constructs a validity matrix. Within every panel-family stratum it selects the first 89 eligible task IDs under a fixed SHA-256 ordering. A task is eligible only when all 27 models completed it and produced a parseable three-item set. The selection code does not load the chosen ingredients or their scores until the task IDs are fixed. This yields a balanced common core without favoring any model or post-selecting easy tasks by observed score.

3.4 Uncertainty and multiplicity

All inferential choices are bound in the content-addressed analysis plan.

- **Scores.** 50000 bootstrap samples are drawn over ingredient anchors; tasks that share an anchor across panels always move together. Each resample is reduced to six panel-family means and then equally weighted. We report pointwise percentile intervals and a simultaneous max- t band across all 27 model scores.
- **Pairs.** Every one of the 351 model pairs uses the same 534 tasks. Two-sided anchor-cluster sign-flip tests use 100000 Monte Carlo draws and Holm correction at familywise $\alpha = .05$. We also report paired mean differences, bootstrap intervals, and Cohen’s d_z .
- **Chance.** Each model is compared with the exact taskwise mean over all 56 portfolios. The 27 paired sign-flip tests form a separate Holm family.
- **Ranks.** Point ranks are accompanied by bootstrap rank intervals and contiguous statistical groups. Models that are not significantly separated are not presented as a resolved total order.

We add a retrospective precision diagnostic that does not alter the confirmatory inference. Within each of the six family-panel strata, we draw 5,000 score-blind subsets without replacement at total task counts 30, 60, 90, 150, and 270, then compare each point order with the complete 534-task order. We report rank correlation, top-five overlap, point-leader preservation, pair-order agreement, and score error. A crossed model-by-task decomposition also reports a descriptive generalizability coefficient for relative endpoint comparisons. It treats the release as a crossed design; it does not turn these fixed endpoints or tasks into random population samples.

The design has 534 independent anchor clusters and 534 paired cells per contrast. Statistical claims are read directly from simultaneous intervals and Holm-adjusted tests; point-rank differences without corrected support are shown as unresolved.

4 Results

4.1 A statistically resolved FlavourBench leaderboard

All 27 endpoints receive a score on the same 534 tasks. Grok 4.6 has the largest point estimate, 65.1, with simultaneous 95% interval [61.0, 69.2]. Across all 351 prespecified paired contrasts, 101 remain significant after Holm correction. A point rank is therefore a compact summary, not a claim that every adjacent model is distinguishable. Figure 2 shows the score scale, simultaneous intervals, and rank groups; Table 3 gives exact values.

The ranked matrix is rectangular: every row contains 534 valid, parser-confirmed responses, so no endpoint is advantaged by answering a different subset. Provider failures and filtered calls remain available in the full execution ledger, but none appears as a zero-valued culinary decision and no incomplete row enters this leaderboard.

4.2 Does the ordering replicate on a second task panel?

The two panel-specific scores use the same frozen metric and 89 tasks per family, but disjoint task IDs. Their model-level Pearson correlation is 0.89 and their rank correlation is 0.80. These are descriptive stability diagnostics, not a model-selection rule; the primary leaderboard and its uncertainty use both panels and cluster shared ingredient anchors. Figure 3 exposes every model rather than reducing replication to a single coefficient.

4.3 How much task evidence is enough for a stable score?

The crossed design’s relative-decision generalizability coefficient is 0.936 at 534 tasks; the same variance model estimates 329 balanced tasks for 0.90. In 5,000 stratified half-size subsamples (270 tasks), the median rank correlation with the complete point order is 0.952 (empirical 95% range 0.897–0.980) and median top-five overlap is 80%. The complete point leader is retained in only 46.1% of those subsets. The first two results show that 534 tasks support a reliable global score; the last shows why a narrow claim about the single point leader would be stronger than the data allow.

4.4 Which pairwise conclusions survive correction?

The score is continuous, and each model pair is compared on the same 534 tasks rather than on a small number of arena votes. Figure 5 visualizes the multiplicity-controlled results. Grey cells are scientifically useful: they mark pairs for which the shared task evidence does not support a directional claim at familywise $\alpha = .05$. This is different from declaring the two models equal.

4.5 Aggregate scores conceal different culinary profiles

Family-level results in Figure 6 show that similar aggregate scores can arise from different strengths. Constraint tasks penalize infeasible portfolios directly, while pairing and substitution reward graded semantic structure. The three columns are related but not interchangeable.

4.6 Rank uncertainty is visible, not hidden

Bootstrap rank intervals expose how often small score differences change order under plausible resampling of ingredient anchors. Figure 7 separates stable regions of the table from visually tempting but statistically unsupported micro-ranks. This is stricter than sorting point estimates and printing an ordinal list without uncertainty.

4.7 Real prompts and model decisions

Table 4 reports one deterministically selected common-core task from each family with responses from the highest- and lowest-point-estimate endpoints. Selection follows a fixed hash rule rather than task score. The public case-study JSON contains the complete prompt, all eight choices, the optimal portfolio, both raw model responses, selected ingredient identities, and exact scores. The examples illustrate why continuous scoring is preferable to exact match: two non-optimal portfolios can differ substantially in culinary utility.

5 From benchmark to training signal

The exhaustive task map makes FLAVOURBENCH useful beyond a static leaderboard. For a task state x_t and selected portfolio a , EPICURE supplies a deterministic reward $r(x_t, a) = s_t(a)$. This creates three immediate training interfaces:

1. supervised learning from the optimal portfolio and the full ordering of 56 candidates;
2. preference or contrastive learning from any pair (a_i, a_j) with known reward difference;
3. offline or online policy optimization using dense bounded rewards and held-out anchors.

The present experiment evaluates models rather than training them. The release therefore keeps a separate, content-addressed lab track: 270 training maps, 72 validation maps, and 84 predeclared transfer maps, balanced across the same six family-panel strata. Every lab anchor is disjoint from the ranked 534-task core and from the other lab splits. Optimizer-facing SFT, DPO, and GRPO views omit the transfer rows. The public maps are not secret, so this is a declared held-out protocol, not a hidden test server. The prospective study fixes two base checkpoints, three training methods, three seeds, and six

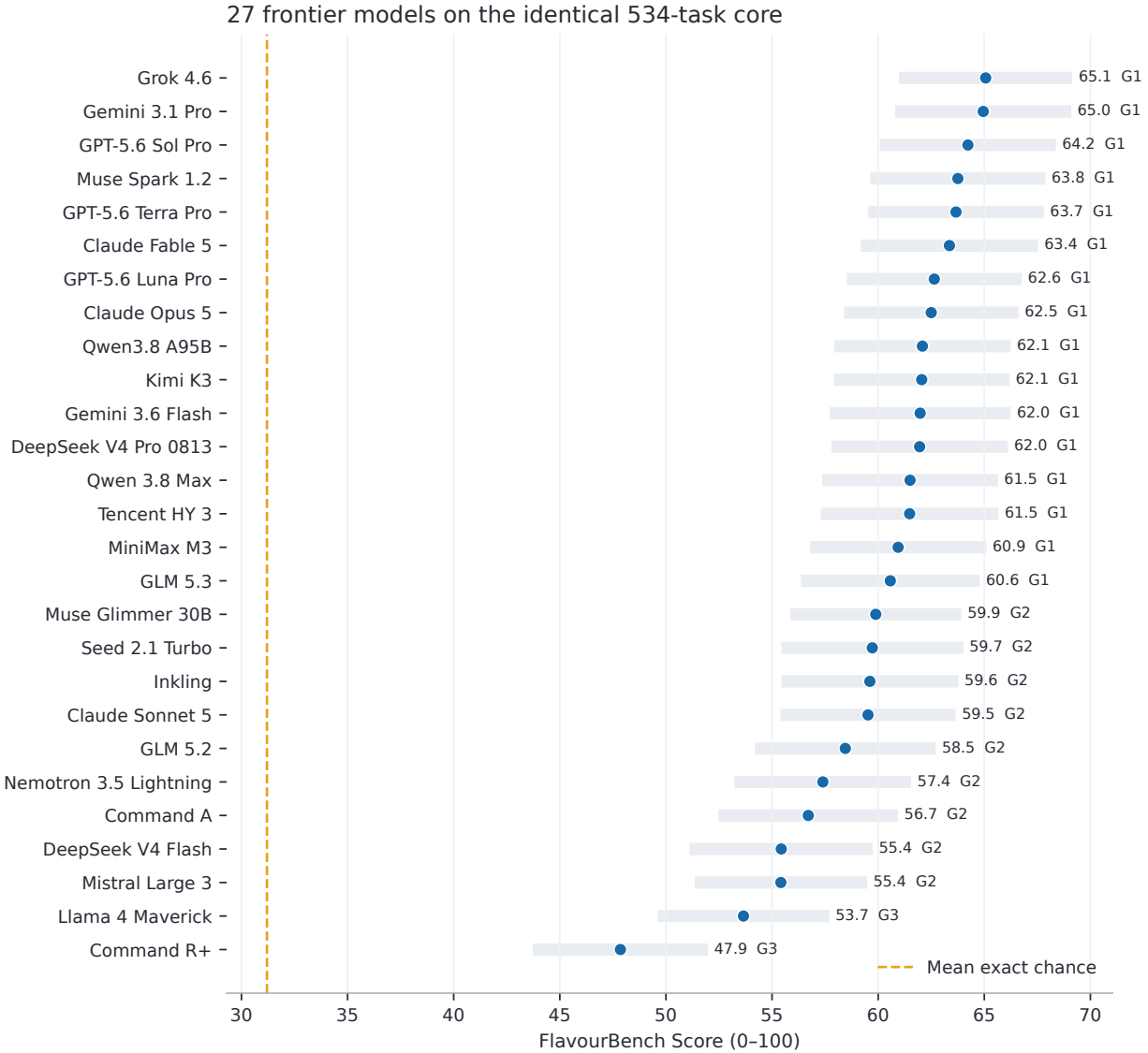


Figure 2: FlavourBench Score with simultaneous 95% max- t intervals. The letter-free group numbers are inferential tiers: a point rank is shown for navigation, while models inside an unresolved group should not be read as a statistically established ordering.

Holm-controlled transfer contrasts before training. Direct optimization on either transfer map would measure memorization rather than culinary learning.

6 Reproducibility

The content-addressed release ([source](#), [dataset](#), and [leaderboard](#)) includes:

- both exact content-addressed task sets (534 scheduled tasks and 534 unique anchors), with prompts, candidates, all 56 portfolio scores, oracle provenance, task strata, and task-set digests;
- the 27-model manifest, exact provider route per model, fallback policy, supported parameters, endpoint contract, and route digest;

- all response records used to construct the 14418-cell ranked matrix, including raw text, completion state, usage, latency, parser output, task score, and digest;
- all content-addressed provider-attempt events referenced by those responses, including normalized and native finish reasons used to distinguish refusals from ordinary completions;
- the frozen joint analysis plan, anchor-cluster contract, seeds, 50,000 bootstrap and 100,000 sign-flip counts, all 351 pairwise rows, cross-panel diagnostics, and the final statistical release; and
- a verifier and asset builder that reconstruct the leaderboard CSV, pairwise CSV, LaTeX tables, and vector figures from local files without provider access.

Responses are written once under a model slot and cell identifier. Existing valid cells are skipped on resume; con-

Rank	Model	FB Score	simultaneous 95% CI	rank 95% CI	group
1	Grok 4.6	65.1	[61.0, 69.2]	[1, 5]	1
2	Gemini 3.1 Pro	65.0	[60.8, 69.1]	[1, 6]	1
3	GPT-5.6 Sol Pro	64.2	[60.1, 68.4]	[1, 8]	1
4	Muse Spark 1.2	63.8	[59.6, 67.9]	[1, 10]	1
5	GPT-5.6 Terra Pro	63.7	[59.5, 67.8]	[1, 11]	1
6	Claude Fable 5	63.4	[59.2, 67.5]	[2, 13]	1
7	GPT-5.6 Luna Pro	62.6	[58.5, 66.8]	[4, 14]	1
8	Claude Opus 5	62.5	[58.4, 66.6]	[3, 15]	1
9	Qwen3.8 A95B	62.1	[57.9, 66.2]	[4, 17]	1
10	Kimi K3	62.1	[57.9, 66.2]	[5, 16]	1
11	Gemini 3.6 Flash	62.0	[57.7, 66.2]	[4, 17]	1
12	DeepSeek V4 Pro 0813	62.0	[57.8, 66.1]	[5, 17]	1
13	Qwen 3.8 Max	61.5	[57.4, 65.6]	[7, 18]	1
14	Tencent HY 3	61.5	[57.3, 65.7]	[6, 19]	1
15	MiniMax M3	60.9	[56.8, 65.1]	[8, 20]	1
16	GLM 5.3	60.6	[56.4, 64.8]	[8, 20]	1
17	Muse Glimmer 30B	59.9	[55.9, 63.9]	[12, 21]	2
18	Seed 2.1 Turbo	59.7	[55.4, 64.0]	[12, 22]	2
19	Inkling	59.6	[55.4, 63.8]	[12, 22]	2
20	Claude Sonnet 5	59.5	[55.4, 63.7]	[13, 22]	2
21	GLM 5.2	58.5	[54.2, 62.7]	[16, 23]	2
22	Nemotron 3.5 Lightning	57.4	[53.2, 61.6]	[19, 25]	2
23	Command A	56.7	[52.5, 60.9]	[20, 25]	2
24	DeepSeek V4 Flash	55.4	[51.1, 59.7]	[22, 26]	2
25	Mistral Large 3	55.4	[51.4, 59.5]	[22, 26]	2
26	Llama 4 Maverick	53.7	[49.6, 57.7]	[24, 26]	3
27	Command R+	47.9	[43.7, 52.0]	[27, 27]	3

Table 3: The complete automated leaderboard. Every score and every pairwise comparison uses the identical 534-task core. Rank intervals come from the anchor-cluster bootstrap; groups summarize the Holm-controlled pairwise graph.

Family	Prompt	Higher-ranked selection	Lower-ranked selection
Substitution	Select three alternatives to ‘boursin cheese’ that best preserve its dairy role, regional context, and mutual portfolio coherence.	caciocavallo, fromage blanc, grana padano (100)	fromage blanc, quail egg, goose egg (15)
Pairing	Select the three-ingredient bundle with the strongest learned pairing to ‘sweetbread’ and the best internal coherence.	parsnip, cipollini onion, porcini mushroom (92)	parsnip, pickled onion, peppadew pepper (0)
Constraints	For a dish centered on ‘potato’, select three vegetarian complements with NOVA processing level at most 2; among valid portfolios, maximize learned pairing and coherence.	parsley, black pepper, thyme (100)	cheese, bay leaf, parsley (0)

Table 4: Three common-core response examples. Parentheses give the fixed task score. The complete prompts and unabridged responses are released in machine-readable form.

flicting artifacts fail closed. Provider fallback is disabled, so the route shown in the manifest is the route measured. The task maps and analysis plan were frozen before the clean primary lineage; transport-calibration responses are retained separately and never enter the leaderboard.

7 Related work

General-purpose evaluation spans broad suites such as HELM [10], difficult knowledge tests such as MMLU-Pro and GPQA [16, 14], live contamination-resistant sets [17, 7], and human preference systems such as Chatbot Arena [2]. Model judges increase scale but introduce their own bias and calibration problems [20]. FLAVOURBENCH instead takes the executable-environment route used by software and agent benchmarks: SWE-bench runs repository tests [8], BFCL evaluates function calls [11], and τ -bench checks interaction outcomes against database state [19].

Culinary datasets and benchmarks cover ingredient

networks, recipe understanding, planning, nutrition, and food knowledge [1, 5, 15, 3, 18, 6, 4, 9]. Our focus is narrower and complementary: common-task frontier model measurement against a versioned culinary decision environment, with exhaustive partial-credit maps and paired inference.

8 Limitations

FlavourBench Score measures agreement with one published EPICURE release, not universal human taste. Within-task min-max normalization makes the 0–100 scale comparable while discarding absolute raw-utility magnitudes. The tasks evaluate constrained ingredient selection rather than full recipe generation, sensory execution, safety advice, or long-horizon kitchen planning. The common-core requirement removes differential missingness from ranking, but it also restricts the estimand to tasks completed by every endpoint; the released supplemental tracks retain the rest. Results bind particular



Figure 3: Panel-1 versus panel-2 FlavourBench Score for every endpoint. The diagonal marks exact agreement. Each point aggregates 89 tasks per family; the joint analysis uses 534 unique anchor clusters.

model routes and collection dates. The dense reward surface supports learning experiments, but this paper does not test whether optimizing EPICURE reward transfers to human cooking outcomes.

9 Conclusion

FLAVOURBENCH makes culinary reasoning measurable without asking one language model to grade another. A versioned runtime enumerates every candidate decision, a 27-model complete panel produces one continuous score, and paired uncertainty distinguishes resolved differences from apparent rank order. The result is an automated leaderboard that can be rerun, audited, and used as a dense training environment. Under this executable culinary ground truth and the released 534-task, 534-cluster distribution, every score, interval, contrast, and figure is reconstructable from public artifacts.

References

- [1] Yong-Yeol Ahn, Sebastian E. Ahnert, James P. Bagrow, and Albert-László Barabási. Flavor Network and the Principles of Food Pairing. *Scientific Reports*, 1:196, 2011.
- [2] Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael I. Jordan,

Joseph E. Gonzalez, and Ion Stoica. Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 8359–8388. PMLR, 2024.

- [3] Donghee Choi, Mogan Gim, Donghyeon Park, Mu-jeen Sung, Hyunjae Kim, Jaewoo Kang, and Jihun Choi. CookingSense: A Culinary Knowledgebase with Multidisciplinary Assertions. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 3983–3996, Torino, Italia, 2024. ELRA and ICCL.
- [4] Tome Eftimov, Ana Gjorgjevikj, Matej Martinc, Gjorgjina Cenikj, Saso Dzeroski, and Barbara Koroušić Seljak. FoodBench-QA: Overview of the Shared Task on Grounded Food and Nutrition Question Answering. In *Proceedings of the Third Workshop on Patient-Oriented Language Processing (CL4Health) at LREC 2026*, 2026.
- [5] Neelansh Garg, Apuroop Sethupathy, Rudraksh Tuwani, et al. FlavorDB: A Database of Flavor Molecules. *Nucleic Acids Research*, 46(D1):D1210–D1216, 2018.
- [6] Andong Hua, Mehak Preet Dhaliwal, Laya Pullela, Ryan Burke, and Yao Qin. NutriBench: A Dataset for Evaluating Large Language Models on Nutrition Estimation from Meal Descriptions. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [7] Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. Live-CodeBench: Holistic and Contamination Free Evaluation of Large Language Models for Code. *arXiv preprint arXiv:2403.07974*, 2024.
- [8] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? In *The Twelfth International Conference on Learning Representations*, 2024.
- [9] Song Jin, Juntian Zhang, Xun Zhang, Zeying Tian, Fei Jiang, Guojun Yin, Wei Lin, Yong Liu, and Rui Yan. DiningBench: A Hierarchical Multi-view Benchmark for Perception and Reasoning in the Dietary Domain. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 35289–35310. Association for Computational Linguistics, 2026.

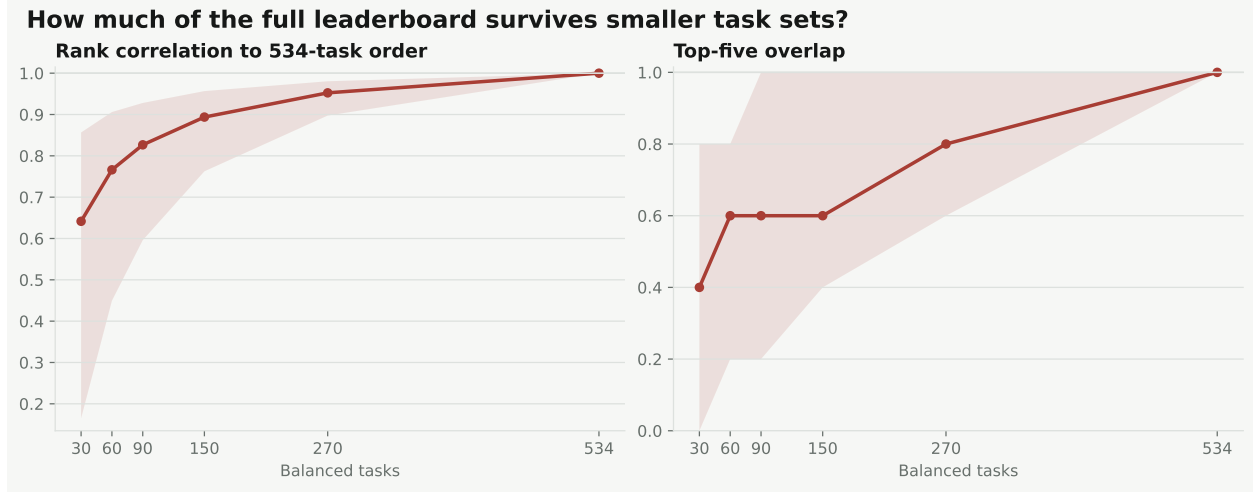


Figure 4: Stability of smaller, score-blind task subsets relative to the complete 534-task point order. Lines are medians across family-panel stratified samples; bands are empirical 2.5th to 97.5th percentiles. The 534-task endpoint is identical to the reference by construction. This is a precision diagnostic, not a post-hoc power calculation.

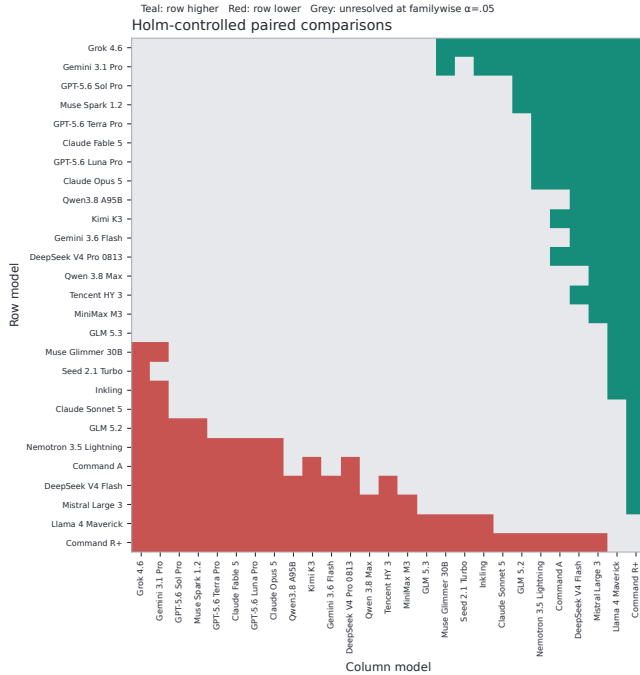


Figure 5: All paired model contrasts after Holm correction. Teal means the row model scores higher; red means lower; grey is unresolved.

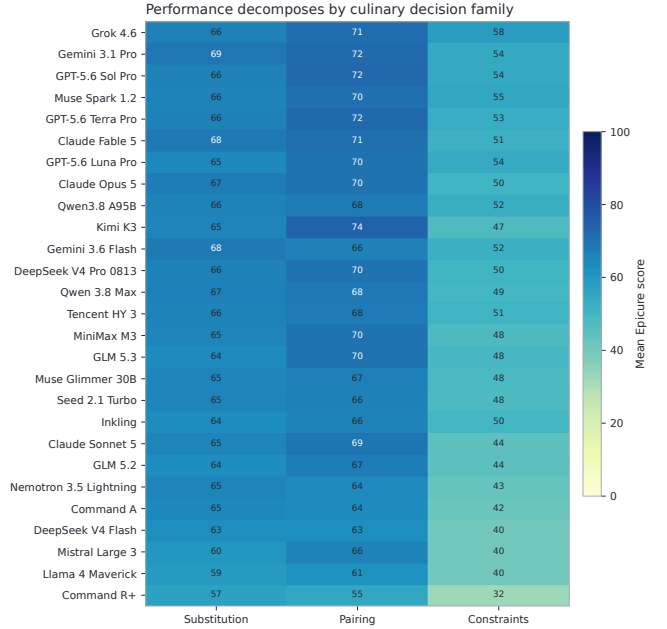


Figure 6: Mean score by family on the common core. Each family contributes 178 tasks per model. Values share the 0–100 scale but retain family-specific decision semantics.

- [10] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic Evaluation of Language Models. *Transactions on Machine Learning Research*, 2023.
- [11] Shishir G. Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and Joseph E. Gonzalez. The Berkeley Function Calling Leaderboard (BFCL): From Tool Use to Agentic Evaluation of Large Language Models. In *Proceedings of the 42nd International Conference on Machine Learning*,

ing, volume 267 of *Proceedings of Machine Learning Research*, pages 48371–48392. PMLR, 2025.

- [12] Jakub Radzikowski and Josef Chen. Epicure: Multidimensional Flavor Structure in Food Ingredient Embeddings. *arXiv preprint arXiv:2604.22776*, 2026.
- [13] Jakub Radzikowski and Josef Chen. Epicure: Navigating the Emergent Geometry of Food Ingredient Embeddings. *arXiv preprint arXiv:2605.22391*, 2026.

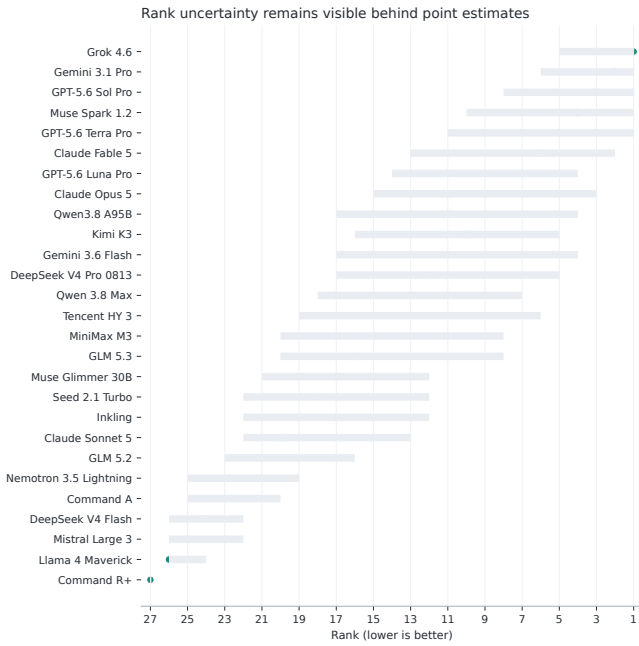


Figure 7: Point rank and 95% bootstrap rank interval for every endpoint. Narrow intervals imply a stable location; overlapping intervals mark unresolved local order.

- [14] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A Graduate-Level Google-Proof Q&A Benchmark. In *First Conference on Language Modeling*, 2024.
- [15] Amaia Salvador, Nicholas Hynes, Yusuf Aytar, Javier Marin, Ferda Ofli, Ingmar Weber, and Antonio Torralba. Learning Cross-Modal Embeddings for Cooking Recipes and Food Images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3020–3028, 2017.
- [16] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- [17] Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, Shubh Agrawal, Sandeep Singh Sandha, Siddhartha Venkat Naidu, Chinmay Hegde, Yann LeCun, Tom Goldstein, Willie Neiswanger, and Micah Goldblum. LiveBench: A Challenging, Contamination-Limited LLM Benchmark. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [18] Zirui Wu, Xiao Liu, Jiayi Li, Lingpeng Kong, and Yansong Feng. Recipe2Plan: Evaluating Planning Abilities of LLMs for Efficient and Feasible Multi-tasking with Time Constraints Between Actions. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 4279–4301, Suzhou, China, 2025. Association for Computational Linguistics.
- [19] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [20] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems*, 36, 2023.

A Exact execution routes

Model	Backend	Panel 1 route	Panel 2 route
Grok 4.6	openrouter	xai/zdr	xai/zdr
Gemini 3.1 Pro	openrouter	google-ai-studio	google-ai-studio
GPT-5.6 Sol Pro	openrouter	openai/flex	openai/flex
Muse Spark 1.2	openrouter	meta	meta
GPT-5.6 Terra Pro	openrouter	openai/flex	openai/flex
Claude Fable 5	openrouter	google-vertex/global	google-vertex/global
GPT-5.6 Luna Pro	openrouter	openai/flex	openai
Claude Opus 5	openrouter	azure/us	azure/us
Qwen3.8 A95B	openrouter	alibaba	alibaba
Kimi K3	openrouter	morph	morph
Gemini 3.6 Flash	openrouter	google-vertex/us	google-vertex/us
DeepSeek V4 Pro 0813	openrouter	gmcloud/fp8	gmcloud/fp8
Qwen 3.8 Max	openrouter	alibaba	alibaba
Tencent HY 3	openrouter	gmcloud/bf16	gmcloud/bf16
MiniMax M3	openrouter	venice/fp8	venice/fp8
GLM 5.3	zai_coding_direct	zai-coding-plan-direct	zai-coding-plan-direct
Muse Glimmer 30B	openrouter	fireworks	fireworks
Seed 2.1 Turbo	openrouter	seed/fp8	seed/fp8
Inkling	openrouter	deepinfra/fp8	deepinfra/fp8
Claude Sonnet 5	openrouter	amazon-bedrock/claude-on-aws	amazon-bedrock/claude-on-aws
GLM 5.2	openrouter	coreweave/fp4	coreweave/fp4
Nemotron 3.5 Lightning	openrouter	venice/fp4	venice/fp4
Command A	openrouter	cohere	cohere
DeepSeek V4 Flash	openrouter	deepinfra/fp4	deepinfra/fp8
Mistral Large 3	openrouter	mistral	mistral
Llama 4 Maverick	openrouter	digitalocean	digitalocean
Command R+	openrouter	cohere	cohere

Table 5: Frozen execution routes by panel. Automatic fallback is disabled; route differences indicate complete, score-blind model-block replacements rather than cell pooling.

B Per-family scores

Model	Substitution	Pairing	Constraints
Grok 4.6	66.1	70.9	58.2
Gemini 3.1 Pro	69.0	71.6	54.2
GPT-5.6 Sol Pro	66.2	72.5	54.1
Muse Spark 1.2	65.9	70.3	55.0
GPT-5.6 Terra Pro	66.1	72.1	52.8
Claude Fable 5	68.2	70.6	51.3
GPT-5.6 Luna Pro	64.5	69.7	53.7
Claude Opus 5	67.2	69.9	50.4
Qwen3.8 A95B	66.0	67.8	52.5
Kimi K3	65.5	73.7	47.0
Gemini 3.6 Flash	68.1	66.2	51.6
DeepSeek V4 Pro 0813	66.4	69.5	49.9
Qwen 3.8 Max	66.6	68.5	49.5
Tencent HY 3	66.0	67.9	50.5
MiniMax M3	65.4	69.9	47.6
GLM 5.3	63.7	69.6	48.4
Muse Glimmer 30B	65.1	66.8	47.8
Seed 2.1 Turbo	65.3	66.2	47.6
Inkling	63.6	65.8	49.5
Claude Sonnet 5	65.3	69.2	44.1
GLM 5.2	63.6	67.2	44.5
Nemotron 3.5 Lightning	65.3	64.3	42.5
Command A	64.7	63.9	41.6
DeepSeek V4 Flash	63.1	63.2	40.1
Mistral Large 3	60.4	65.8	40.0
Llama 4 Maverick	59.4	61.4	40.2
Command R+	56.7	54.9	32.0

Table 6: Mean score in each 178-task family. Every entry uses the same tasks for every model; the equal-family mean is the FlavourBench Score.

C Protocol constants

Quantity	Frozen value
Models	27
Collection panels	2
Primary tasks per model	534
Unique anchor clusters	534
Tasks per family	178
Candidate ingredients	8
Selected ingredients	3
Scored portfolios per task	56
Complete response cells	14418
Bootstrap replicates	50000
Sign-flip replicates	100000
Pairwise hypotheses	351
Shared cells per pair	534
Common-core rule	Status/parser only; score-blind

Table 7: Core design and inference constants fixed before clean primary execution.